

RESEARCH INTERESTS

AI Efficiency (Quantization, Parameter-Efficient Fine-Tuning, Lossless Compression); AI Agents (Self-Evolving, Long-Horizon Planning, Multi-Agent Collaboration); AI Safety (Fairness, Privacy)

EDUCATION

Rice University Ph.D. in Electrical and Computer Engineering, Advisor: Prof. Anshumali Shrivastava	August 2024 – Present
Rice University M.S. in Data Science, Advisor: Prof. Joe Warren and Prof. Akane Sano	August 2021 – May 2023
University of Electronic Science and Technology of China B.E. in Electronic Information Engineering, GPA: 3.66/4.00	August 2017 – May 2021

INDUSTRY EXPERIENCE

Meta Superintelligence Lab <i>Visiting Researcher</i> , with Anshumali Shrivastava and Jason Chen	August 2025 – Present
• [ARXIV’26] Developed SIRA, a training-free retrieval agent that compresses multi-round search into a single corpus-discriminative BM25 call by using an LLM to expand documents and queries with evidence vocabulary filtered by corpus document-frequency statistics, achieving the best average performance across ten BEIR benchmarks over dense, learned-sparse, and LLM search-agent baselines [8]. • [ARXIV’26] Constructed BrowseComp-Wikipedia, a new hard-search benchmark of 232 queries grounded in a 25.6M-document Wikipedia index, on which SIRA outperforms state-of-the-art multi-round agents using Perplexity search at every retrieval budget [8].	

RESEARCH EXPERIENCE

Rice Rush Lab <i>Graduate Research Assistant</i> , with Prof. Anshumali Shrivastava	December 2024 – Present
• [ICLR’26] Developed an entropy-based lossless compression algorithm for float8 neural network parameters, achieving an 18% compression ratio while maintaining bit-exact reconstruction of the original model [3]. • [ICML’26] Contributed to Sketch&Walk Attention, a training-free sparse attention method that uses Hadamard sketching to approximate attention scores and a cross-layer walk mechanism to select top-k attention blocks, achieving up to 6× inference speedup while maintaining near-lossless accuracy at 20% attention density [2]. • Proposed a compressed adaptation framework for diffusion transformers that performs global clustering of weights and represents each weight via a global cluster ID and an index into that cluster’s lookup table, matching or exceeding full-precision LoRA while substantially outperforming QLoRA in quality, training time, and memory across 6 diffusion transformers (2B–20B) [1].	
Rice Computational Wellbeing Group <i>Graduate Student Researcher</i> , with Prof. Akane Sano	January 2024 – July 2024
• [TMLR’25] Developed a tabular diffusion model with sensitive guidance that generates fair synthetic data with balanced joint distributions of target labels and sensitive attributes, outperforming existing methods on fairness metrics such as demographic parity ratio and equalized odds ratio by over 10% [7]. • [CHIL’25] Designed a self-supervised contrastive pretraining method that jointly learns from multimodal wearable sensor time series and survey data, improving stress detection accuracy while reducing reliance on labeled data [4].	

Graduate Student Researcher, with Prof. Joe Warren

- [GITHUB'22] Developed a Python toolbox for animating the training progress of multi-layer perceptrons [10].

Monash Data Futures Institute

December 2020 – March 2021

Undergraduate Student Researcher, with Prof. Hao Wang

- [ARXIV'23] Developed an asynchronous distributed alternating direction method of multipliers (ADMM) algorithm to optimize energy trading problems under asynchronous communication, allowing communication delay and indicating a potential for better outcomes in real-world applications [9].

SELECTED PUBLICATIONS[\[FULL LIST\]](#)*In Submission:*

- [1] **Zeyu Yang**, Yide Ran, Tianyi Zhang, Hoang Anh Duy Le, Sahil Joshi, Zhaozhuo Xu, Anshumali Shrivastava, “GSS: Compression and Extreme Parameter-Efficient Fine-Tuning of Diffusion Transformers via Globally-Sensitive Sketch”, submitted to *Neural Information Processing Systems (NeurIPS)*, 2026.

Conference Publications:

- [2] Hoang Anh Duy Le, Sahil Joshi, **Zeyu Yang**, Zhaozhuo Xu, Anshumali Shrivastava, “Scout Before You Attend: Sketch-and-Walk Sparse Attention for Efficient LLM Inference”, in *International Conference on Machine Learning (ICML)*, 2026.
- [3] **Zeyu Yang**, Tianyi Zhang, Jianwen Xie, Chuan Li, Zhaozhuo Xu, Anshumali Shrivastava, “To Compress or Not? Pushing the Frontier of Lossless GenAI Model Weights Compression with Exponent Concentration”, in *International Conference on Learning Representations (ICLR)*, 2026.
- [4] **Zeyu Yang**, Han Yu, Akane Sano, “Contrastive Pretraining for Stress Detection with Multimodal Wearable Sensor Data and Surveys”, in *Conference on Health, Inference, and Learning (CHIL)*, 2025.
- [5] Yuanhao Gong, Tan Tang, **Zeyu Yang**, Lantao Yu, “A Filter for Minimizing Gaussian Curvature on 3D Triangular Meshes”, in *International Symposium on Biomedical Imaging (ISBI)*, 2025.
- [6] Yizhuo Yang, Huan Wang, Zhiliang Liu, **Zeyu Yang**, “Few-Shot Learning for Rolling Bearing Fault Diagnosis via Siamese Two-Dimensional Convolutional Neural Network”, in *Asia-Pacific International Symposium on Advanced Reliability and Maintenance Modeling (PHM)*, 2020.

Journal Publications:

- [7] **Zeyu Yang**, Han Yu, Peikun Guo, Khadija Zanna, Xiaoxue Yang, Akane Sano, “Balanced Mixed-Type Tabular Data Synthesis with Diffusion Models”, *Transactions on Machine Learning Research (TMLR)*, 2025.

Preprints:

- [8] **Zeyu Yang**, Qi Ma, Jason Chen, Anshumali Shrivastava, “Superintelligent Retrieval Agent: The Next Frontier of Agentic Retrieval”, *arXiv:2605.06647*, 2026.
- [9] **Zeyu Yang**, Hao Wang, “Network-Aware Asynchronous Distributed ADMM Algorithm for Peer-to-Peer Energy Trading”, *arXiv:2312.06976*, 2023.

Published Software:

- [10] *PlotNet*. (2022). [Online]. Available: <https://github.com/zeyuyang8/plotnet>

AWARDS & HONORS

Outstanding Undergraduate Student Award	2021
Meritorious Winner of Interdisciplinary Contest in Modeling	2020

SKILLS**Programming:**

- *Languages*: C++, Rust, CUDA, Python, JavaScript, MATLAB, Shell Scripting
- *ML & Inference Frameworks*: PyTorch, Hugging Face, vLLM, SGLang, DeepSpeed, Weights & Biases

- *Distributed Training*: FSDP, Megatron-LM, verl, Ray, Accelerate, NCCL, Tensor/Pipeline/Data Parallelism
- *Web Development*: Django, Dash, React, HTML, CSS, PostgreSQL, Docusaurus
- *Tools & Infrastructure*: Git, Docker, AWS, Linux, LaTeX, Markdown, Tableau, Claude Code

Machine Learning:

- *Deep Learning*: Large-Language Models, Diffusion Models, Transformers, Contrastive Learning
- *Methods*: Information Retrieval, BM25, Quantization, Sparse Dictionary Learning, Entropy Coding, Matrix Factorization, Feature Selection, Bias Mitigation

COURSEWORK

Graduate Coursework:

- *Computer Science*: Programming for Data Science, Big Data Management for Data Science, Data Visualization, Graduate Design and Analysis of Algorithms
- *Machine Learning*: Statistical Machine Learning, Deep Learning for Vision and Language, Applied Machine Learning and Data Science Projects, AI for Health, Advanced Machine Learning, Learning from Sensor Data
- *Statistics & Optimization*: Statistics for Data Science, Convex Optimization, Information Theory

Undergraduate Coursework:

- *Computer Science*: Introductory C Programming, Introductory Python Programming
- *Electrical Engineering*: Application and Design of Digital Logic, Signals and Systems, Digital Communication, Micro-electronic Systems, Circuit Analysis and Design, Fundamentals of Analog Circuits
- *Math & Physics*: Linear Algebra and Space Analytic Geometry, Probability Theory and Mathematical Statistics, Calculus I, Calculus II, Physics I, Physics II

TEACHING EXPERIENCE

Introduction to Data Science (DSCI 101) at Rice University, <i>Teaching Assistant</i> , with Prof. Lorenzo Luzi	Fall 2025
Data Visualization (COMP 665) at Rice University, <i>Teaching Assistant</i> , with Prof. Joe Warren	Spring 2022 – Spring 2023
Statistics for Data Science (COMP 680) at Rice University, <i>Academic Tutor</i> , with Prof. Chen Su	Spring 2023